altitude, speed, or orientation. Without data transmission, adjustments were made purely by observation, resulting in delayed reactions and inaccurate flight control.

- *Limited military and commercial utility*
  Although they demonstrated the feasibility of unmanned flight, first-generation drones lacked the technological maturity for large-scale military or commercial application. Their use was confined mainly to anti-aircraft target practice and basic experimental purposes, offering limited tactical or operational value.
- *Technological immaturity and high development costs*
  The technology of the early 20th century was not advanced enough to support reliable UAV systems. Components such as radio transmitters, lightweight engines, and control servos were expensive and difficult to produce in quantity. As a result, most projects remained experimental prototypes rather than deployable systems.

2.8 Case Study: Early Drone Use in Military Exercises

The use of first-generation drones in early military exercises marked a pivotal moment in defense innovation, showcasing the potential of unmanned aerial vehicles (UAVs) as practical tools for training, reconnaissance, and experimentation. Although limited in sophistication, these early systems demonstrated that remotely operated aircraft could enhance military preparedness, improve targeting accuracy, and reduce risks to human pilots.

One of the most notable examples comes from the United Kingdom in the mid-1930s, with the deployment of the De Havilland DH.82B *Queen Bee* during large-scale anti-aircraft training operations. Developed under the direction of the British Air Ministry, the Queen Bee was a radio-controlled version of the Tiger Moth biplane, designed to simulate enemy aircraft for